

State of the Art in Full Waveform Inversion

by Ian F. Jones

Over the past ten years, full waveform inversion (FWI) has emerged and developed to the point that is now the main technique of choice for detailed model building and reflectivity estimation for complex geological environments.

In this review, I will outline the underlying principles involved in FWI, detailing the use of RTM to estimate the location of subsurface parameter error, and introduce the many and varied 'flavors' of FWI, noting their limitations and benefits.

Use of FWI for model estimation, reflectivity generation, and pre-sack attribute analysis (AVA) will be covered, and demonstrated with field data examples.

Duration

One day.

Intended Audience

Entry level

Prerequisites (Knowledge/Experience/Education Required)

The course is designed to be followed by anyone with a broad geoscience background: no specific detailed foreknowledge is required, although a familiarity with geophysical terminology will be useful.

Course outline

For the topics listed below, real data examples will be used to demonstrate the application and limitation of each technique.

- Why do we need a detailed velocity model?
- Migration using wavefield extrapolation methods (WEM, RTM, etc.)
 - One-way versus two-way propagation
- Resolving short-scale-length velocity anomalies:
- The mechanics of tomographic inversion with wavefield extrapolation theory (FWI)
 - Refraction versus reflection
 - Different 'norms' (dynamic versus kinematic: least-squares, travel time, phase, optimal transport)
- Examples of current industrial practice for various geological settings:

Learner Outcomes

Upon completion of this course, the participants should be able to:

1. Describe how wavefield extrapolation migration works, in terms of the underlying physics and the associated approximations involved.

2. Characterize the limitations of differing FWI schemes, in terms of:
 1. the components of the wavefield being used,
 2. the (minimization) norms being used,
 3. use of multiple and ghost energy,
 4. reflectivity estimation.
 5. AVA applications.